

Java 7 - Quick Sort - Detyra të zgjidhura & detyra për ushtrime

Lënda: Hyrje në Algoritme

Detyra 1: Te sortohet array a[ ]= 7, 5, 1, 3, 6, 4, 2 me quicksort Sedgewick 2 way partitioning.
Shuffled array: **a[ ] = 5,3,1,7,4,2,6**

5 3 1 7 4 2 6

Zgjidhje:

5 3 1 7 4 2 6 | **1)** a[i] < a[lo]
lo i j | 3 < 5 (Po), i++

5 3 1 7 4 2 6 | **1)** a[i] < a[lo]
lo i j | 1 < 5 (Po), i++

5 3 1 7 4 2 6 | **1)** a[i] < a[lo] **2)** a[j] > a[lo]
lo i j | 7 < 5 (Jo), 6 > 5 (Po), j--

5 3 1 7 4 2 6 | **2)** a[j] > a[lo] **3)** exchange a[i] me a[j]
lo i j | 2 > 5 (Jo),

5 3 1 2 4 7 6 | **1)** a[i] < a[lo]
lo i j | 2 < 5 (Po), i++

5 3 1 2 4 7 6 | **1)** a[i] < a[lo]
lo i j | 4 < 5 (Po), i++

5 3 1 2 4 7 6 | **1)** a[i] < a[lo] **2)** a[j] > a[lo]
lo i, j | 7 < 5 (Jo), 7 > 5(Po), j--

5 3 1 2 4 7 6 | i and j crossed, exchange a[lo] with a [j]
lo j i |

4 3 1 2 **5** 7 6 | **first pivot**

4 3 1 2 | **1)** a[i] < a [lo]
lo **i** **j** | 3 < 4 (Po) i++

4 3 1 2 | **1)** a[i] < a [lo]
lo **i** **j** | 1 < 4 (Po) i++

4 3 1 2 | **1)** a[i] < a [lo]
lo **i,j** | 2 < 4 (Po) i++

4 3 1 2 | i and j crossed, exchange a[lo] with a [j]
lo **j** **i** |

2 3 1 **4** | **second pivot**

Detyra 2: Te sortohet array a[ ] = 7,3,2,6,5,8,4 me quicksort Sedgewick 2 way partitioning.
Shuffled array: **a[ ] = 6,3,7,2,8,5,4**

6 3 7 2 8 5 4

Zgjidhje:

6 3 7 2 8 5 4 | **1)** a[i] < a[lo]
lo **i** **j** | 3 < 6 (Po), i++

6 3 7 2 8 5 4 | **1)** a[i] < a[lo] **2)** a[j] > a[lo] **3)** exchange a[i] me a[j]
lo **i** **j** | 7 < 6 (Jo), 4 > 6 (Jo)

6 3 4 2 8 5 7 | **1)** a[i] < a[lo]
lo **i** **j** | 4 < 6 (Po), i++

6 3 4 2 8 5 7 | **1)** a[i] < a[lo]
lo **i** **j** | 2 < 6 (Po), i++

6 3 4 2 8 5 7 | **1)** a[i] < a[lo] **2)** a[j] > a[lo]
lo **i** **j** | 8 < 6 (Jo), 7 > 6 (Po), j--

6 3 4 2 8 5 7 | **2)** a[j] > a[lo] **3)** exchange a[i] me a[j]
lo **i** **j** | 5 > 6 (Jo),

6 3 4 2 5 8 7 | **1)** a[i] < a[lo]
lo **i** **j** | 5 < 6 (Po), i++

6 3 4 2 5 8 7 | **1)** a[i] < a[lo] **2)** a[j] > a[lo]
lo **i,j** | 8 < 6 (Jo), 8 > 6 (Po), j--

6 3 4 2 5 8 7 | i and j crossed, exchange a[lo] with a [j]
lo **j** **i** |

5 3 4 2 **6** 8 7 | **first pivot**

5 3 4 2 | **1)** a[i] < a[lo]
lo **i** **j** | 3 < 5 (Po), i++

5 3 4 2 | **1)** a[i] < a[lo]
lo **i** **j** | 4 < 5 (Po), i++

5 3 4 2 | **1)** a[i] < a[lo]
lo **i,j** | 2 < 5 (Po), i++

5 3 4 2 | i and j crossed, exchange a[lo] with a [j]
lo **j** **i** |

2 3 4 **5** | **second pivot**

Detyra 3. Te sortohet array a[ ] = 3,2,7,5,4,3,3,2 me quicksort Dijkstra 3 - way partitioning.
Shuffled array: **a[ ] = 3,5,2,4,7,3,3,2**

3 5 2 4 7 3 3 2

Zgjidhje:

3	**5**	**2**	**4**	**7**	**3**	**3**	**2**	1) 3 < 3 (Jo), 2) 3 > 3 (Jo), 3) 3==3 (Po) i++
lt,i							**gt**	
3	**5**	**2**	**4**	**7**	**3**	**3**	**2**	1) 5 < 3 (Jo), 2) 5 > 3 (Po) exchange a[gt] with a[i], gt--
lt	**i**						**gt**	
3	**2**	**2**	**4**	**7**	**3**	**3**	**5**	1) 2 < 3 (Po), exchange a[lt] with a[i], i++, lt++
lt	**i**					**gt**		
2	**3**	**2**	**4**	**7**	**3**	**3**	**5**	1) 2 < 3 (Po), exchange a[lt] with a[i], i++, lt++
	lt	**i**				**gt**		
2	**2**	**3**	**4**	**7**	**3**	**3**	**5**	1) 4 < 3 (Jo), 2) 4 > 3 (Po) exchange a[gt] with a[i], gt--
		lt	**i**			**gt**		
2	**2**	**3**	**3**	**7**	**3**	**4**	**5**	1) 3 < 3 (Jo), 2) 3 > 3 (Jo), 3) 3==3 (Po) i++
		lt	**i**		**gt**			
2	**2**	**3**	**3**	**7**	**3**	**4**	**5**	1) 7 < 3 (Jo), 2) 7 > 3 (Po) exchange a[gt] with a[i], gt--
		lt		**i**	**gt**			
2	**2**	**3**	**3**	**3**	**7**	**4**	**5**	1) 3 < 3 (Jo), 2) 3 > 3 (Jo), 3) 3==3 (Po) i++
		lt		**i,gt**				
2	**2**	**3**	**3**	**3**	**7**	**4**	**5**	i and gt cross, gjendet first pivot
		lt		**gt**	**i**			
2	**2**	**3**	**3**	**3**	**7**	**4**	**5**	**first pivot**

2 **2** | 1) 2 < 2 (Jo), 2) 2 > 2 (Jo), 3) 2==2 (Po) i++

lt, i **gt** |

2 **2** | 1) 2 < 2 (Jo), 2) 2 > 2 (Jo), 3) 2==2 (Po) i++

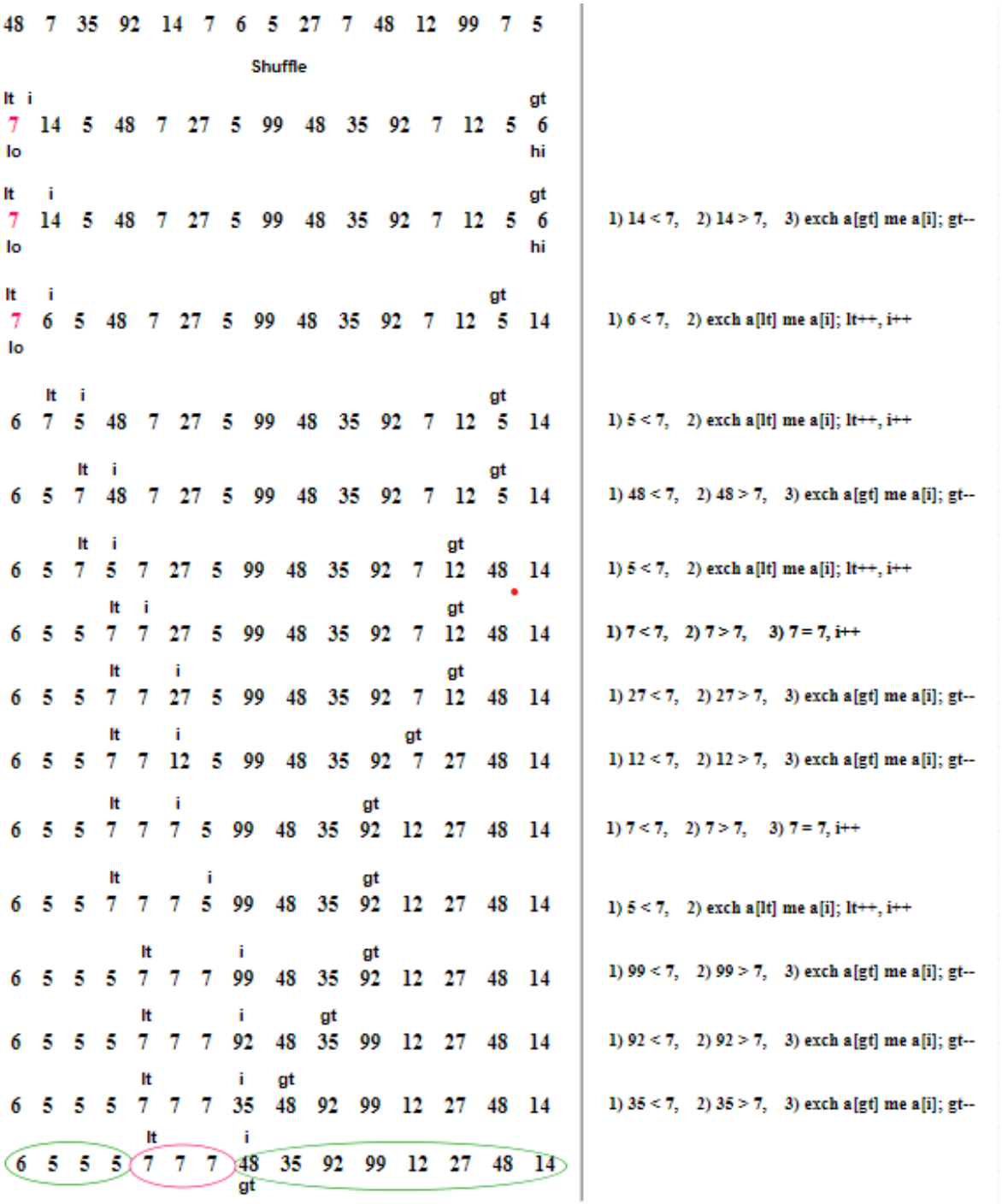
lt **i,gt** |

2 **2** | i and gt cross, gjendet second pivot

lt **gt** **i** |

Detyra 3. Të sortohet vargu a[] = 48, 7, 35, 92, 14, 7, 6, 5, 27, 7, 48, 12, 99, 7, 5 duke përdorur algoritmin Quicksort - Dijkstra 3-way:

48 7 35 92 14 7 6 5 27 7 48 12 99 7 5



6 5 5 5 | 1) 5 < 6 (Po), exchange a[lt] with a[i], i++, lt++

lt, i gt |

5 6 5 5 | 1) 5 < 6 (Po), exchange a[lt] with a[i], i++, lt++

lt i gt |

5 5 6 5 | 1) 5 < 6 (Po), exchange a[lt] with a[i], i++, lt++

lt i,gt |

5 5 5 6 | 1) 5 < 6 (Po), exchange a[lt] with a[i], i++, lt++

lt,gt i | i and gt cross, **pivot 2 found**

Detyra për ushtrime

Detyra 1. Të sortohet vargu duke përdorur algoritmin Quick Sort - Sedgewick 2-way partitioning:

3 24 5 8 5 9 4 1 2 6

Detyra 2. Të sortohet vargu duke përdorur algoritmin Quick Sort - Sedgewick 2-way partitioning:

Q U I C K S O R T

Detyra 3. Të sortohet vargu duke përdorur algoritmin Quicksort - Dijkstra 3-way

P A B X W P P V P D P C Y Z

Detyra 4. Të sortohet vargu duke përdorur algoritmin Quicksort - Dijkstra 3-way

6 7 9 5 9 5 7 1 2 9